`   CO1 ASSESSMENT TOOL -1

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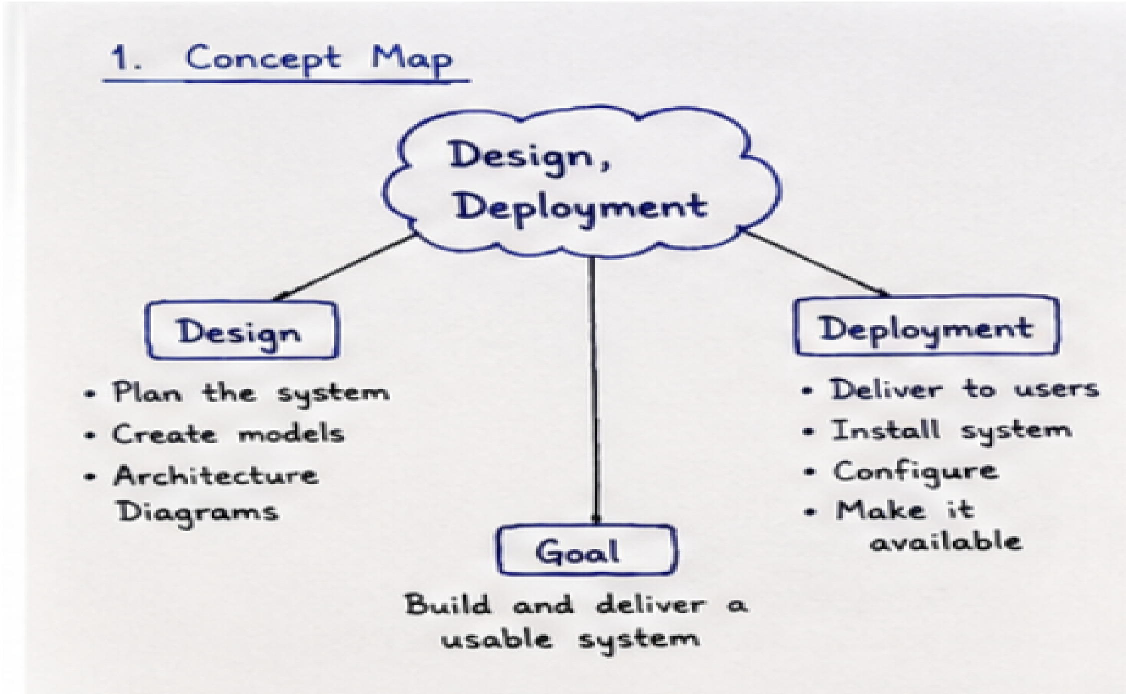
COURSE:Object Oriented Analysis And Design
CODE:0307

1.**Explanation**

Rules of inference are logical rules used to derive a valid conclusion from given premises.

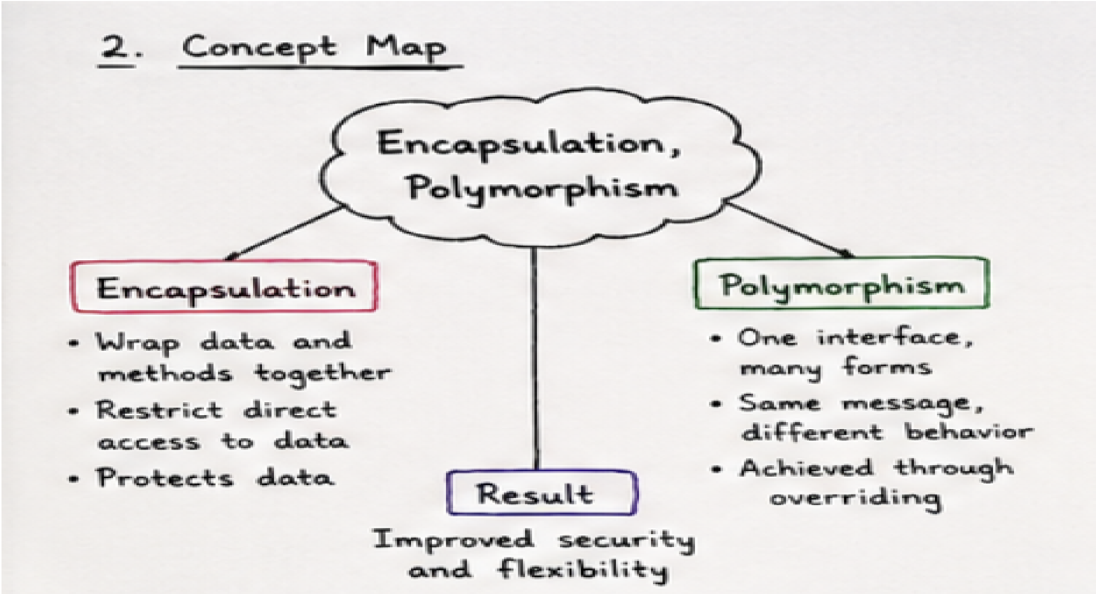


Analysis

1. **Identify the premises** given in the question.
2. **Identify the conclusion** that needs to be proved.
3. Match the premises with a suitable rule of inference.
4. Apply the rule and derive an intermediate statement.
5. Continue applying valid inference rules until the required conclusion is obtained.
6. If the conclusion is obtained logically from the premises, the argument is **valid**.
7. If it cannot be derived using valid inference steps, the argument is **not valid**.

2.Explanation

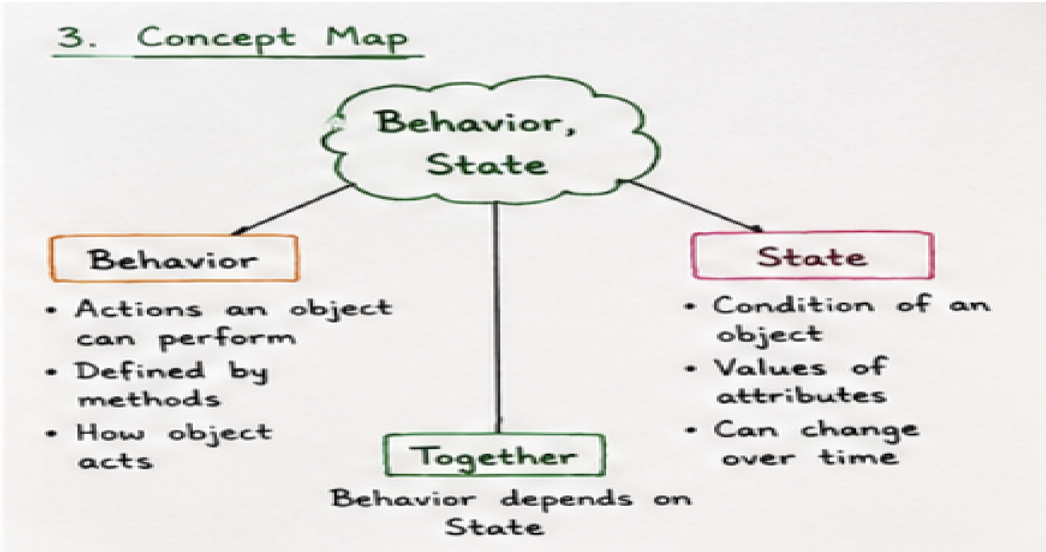
Encapsulation means binding data and methods together, while **polymorphism** allows one interface to have different implementations.



Analysis

1. Identify the **class and its data**.
2. Encapsulate data using **methods**.
3. Identify the common **interface/method**.
4. Apply **polymorphism** for different behaviours.
5. Check whether the design is **flexible and reusable**.

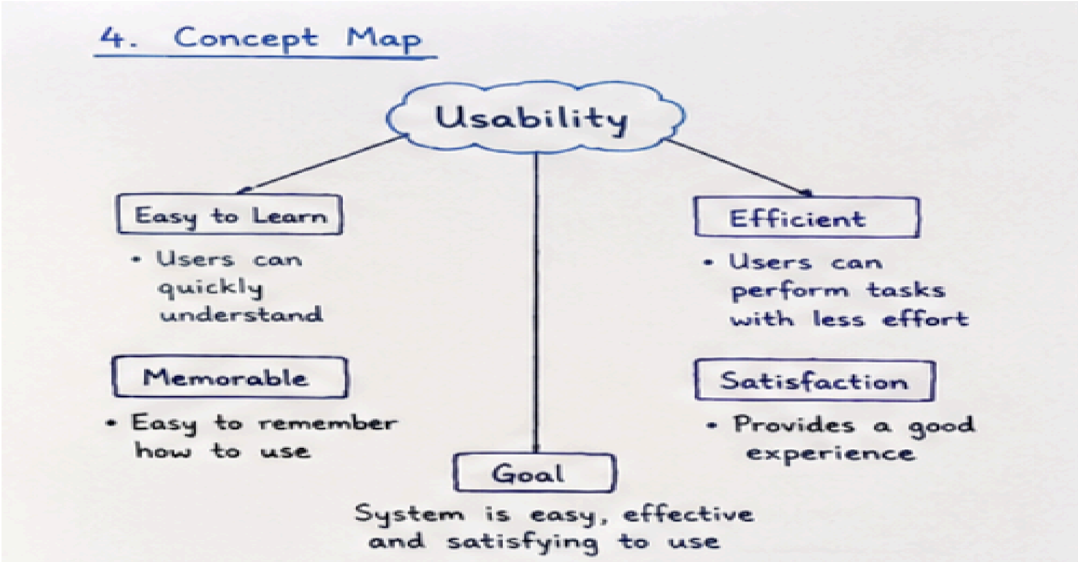
3.Explanation
PCNF (Principal Conjunctive Normal Form) is a standard form of a compound proposition written as an **AND (∧) of maxterms**.



Analysis

1. Identify the given **compound proposition**.
2. Apply **logical equivalences** step by step.
3. Convert the expression into **CNF form**.
4. Express it as **principal maxterms**.
5. The final expression is the required **PCNF**.

4.Explanation
Rules of inference help us derive a valid conclusion from given statements or premises. The conclusion is accepted when it logically follows from the premises.



Analysis

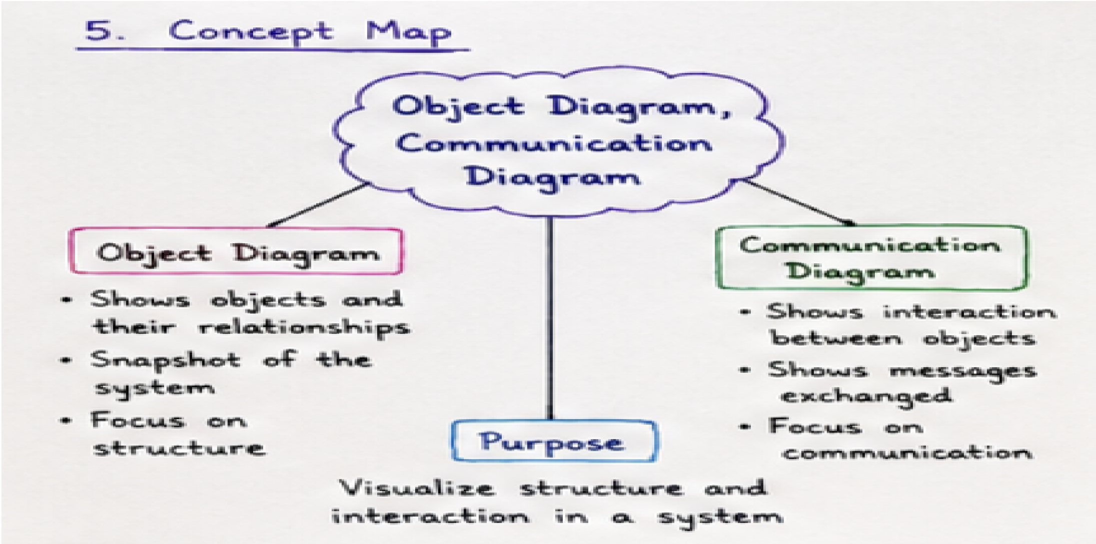
1. Identify the **given premises**.
2. Identify the **required conclusion**.
3. Select the suitable **rule of inference**.
4. Apply the rule to derive the conclusion.
5. Verify that the conclusion is **logically valid**.

5.Explanation

An equivalence relation is a relation that satisfies reflexive, symmetric, and transitive properties. If all three are satisfied, the relation is an equivalence relation.

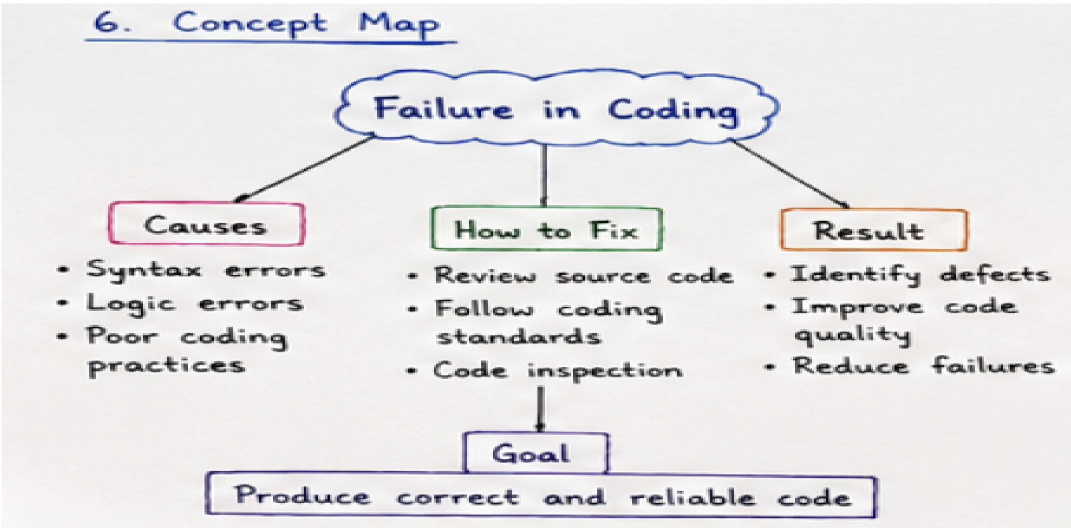
Analysis

1. Identify the **set and relation** given.
2. Check the **reflexive** property.
3. Check the **symmetric** property.
4. Check the **transitive** property.
5. If all three hold, conclude that it is an **equivalence relation**.



6.Explanation

Failure in Coding means errors introduced while writing the program, which can affect the correctness and quality of the code.

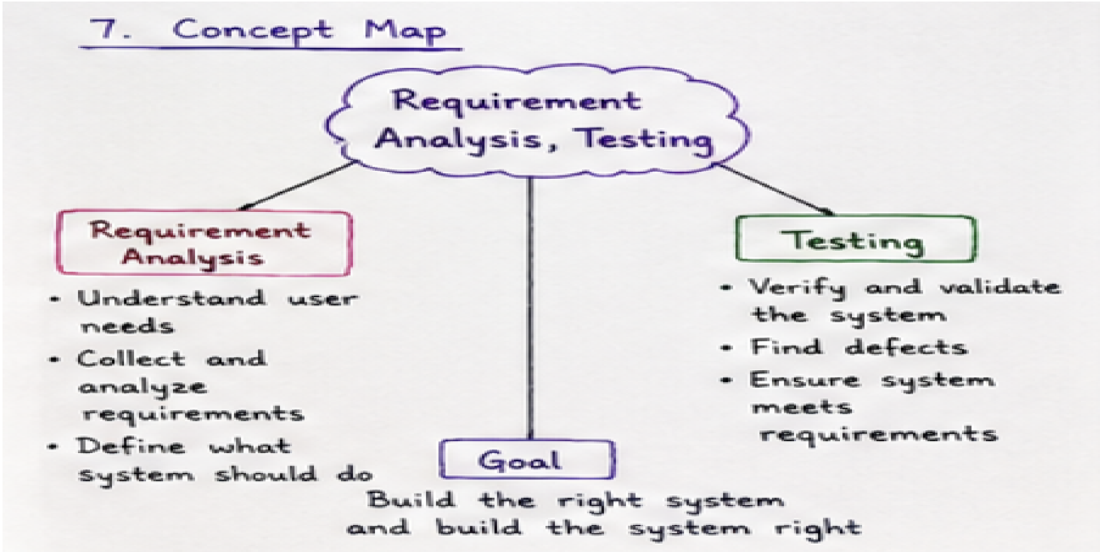


Analysis

1. Identify **syntax, logic, and coding errors**.
2. Review the **source code** carefully.
3. Follow proper **coding standards**.
4. Perform **code inspection**.
5. Aim to produce **correct and reliable code**.

7.Explanation

Requirement Analysis and Testing help to build the right system and ensure that the system meets its requirements.



Analysis

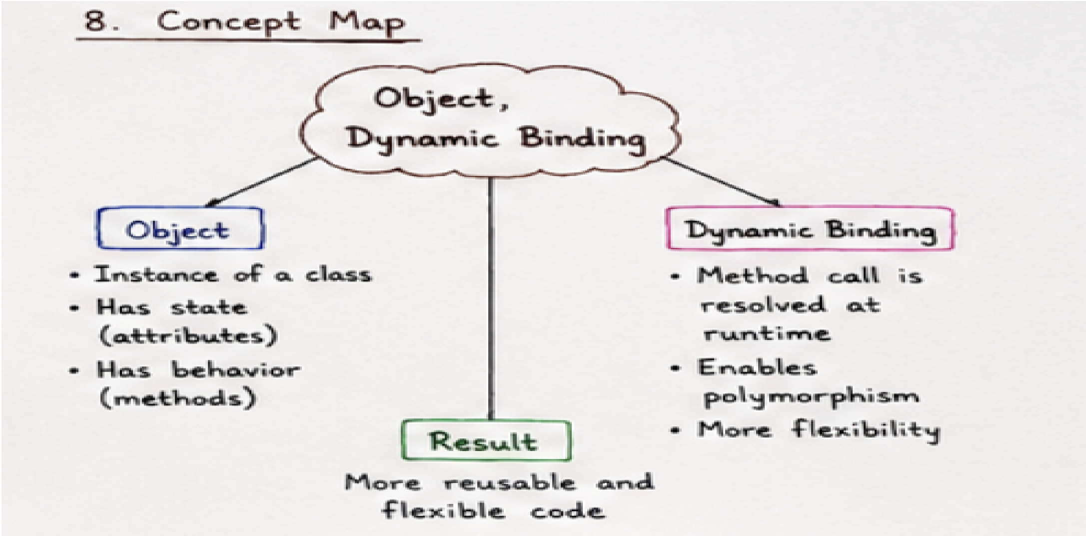
1. Understand the **user needs**.
2. Collect and analyze **requirements**.
3. Define what the **system should do**.
4. **Verify and validate** the system.
5. Find defects and ensure the system **meets requirements**.

8.Explanation

Object is an instance of a class. **Dynamic Binding** resolves a method call at runtime and supports polymorphism.

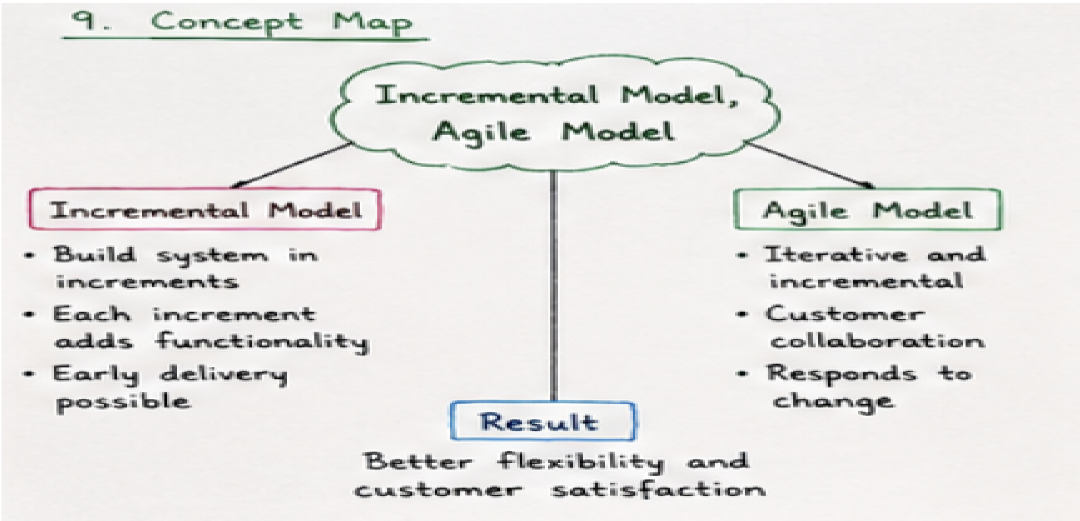
Analysis

1. Identify the **object**.
2. Identify its **state and behavior**.
3. Understand the **method call**.
4. Resolve the method at **runtime**.
5. Achieve **flexible and reusable code**.



9.Explanation

Incremental Model develops the system in small increments. **Agile Model** uses iterative development and responds to changes.

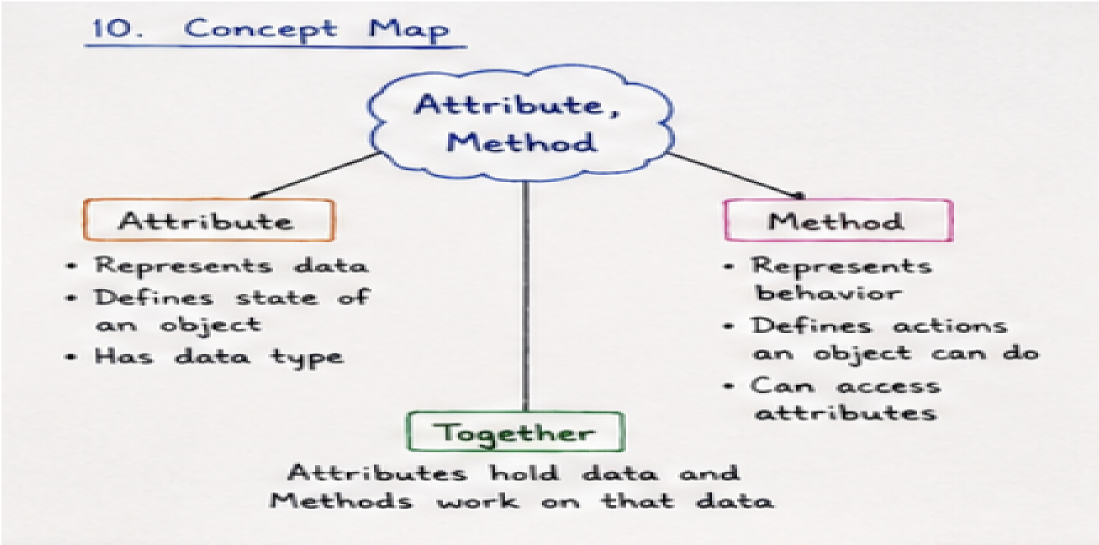


Analysis

1. Divide the system into **increments**.
2. Build and deliver each increment.
3. Use **iterative development**.
4. Encourage **customer collaboration**.
5. Respond to changes and improve **customer satisfaction**.

10.Explanation

An **Attribute** represents the data or state of an object.
A **Method** represents the behavior or actions of an object.

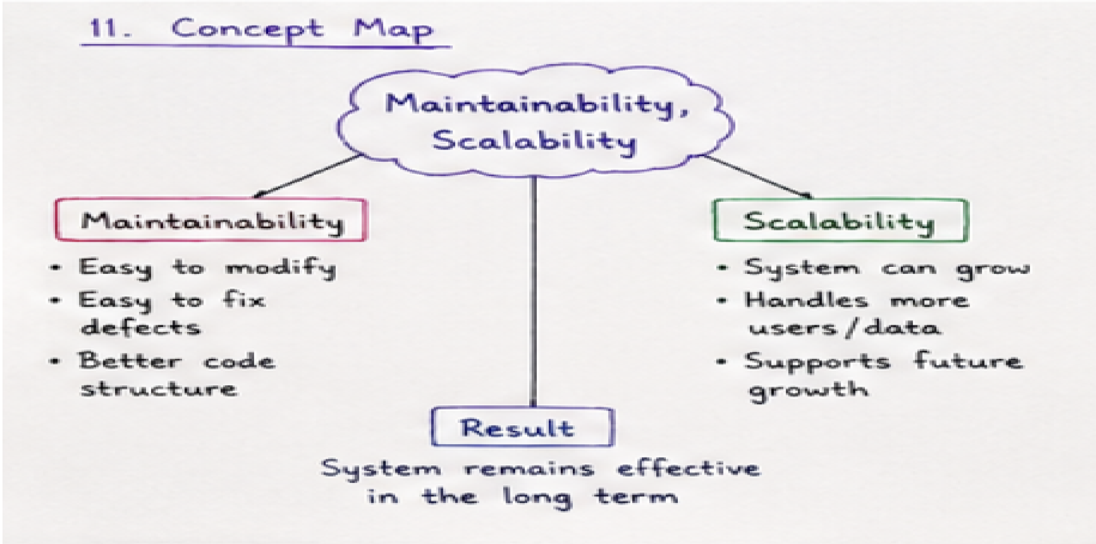


Analysis

1. Identify the object's **data**.
2. Define its **attributes**.
3. Identify the object's **behavior**.
4. Define its **methods**.
5. Attributes hold data and methods **work on that data**.

11.Explanation

Maintainability makes a system easy to modify and fix. **Scalability** allows the system to grow and handle more users or data.



Analysis

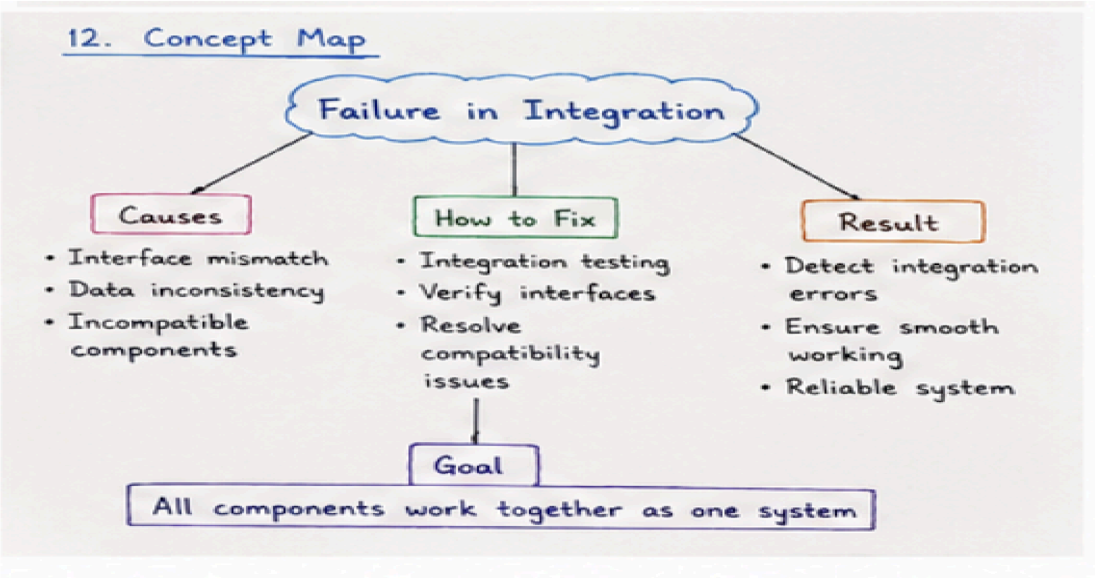
1. Make the system easy to **modify**.
2. Make defects easy to **fix**.
3. Maintain a better **code structure**.
4. Support increasing **users and data**.
5. Ensure the system remains effective in the **long term**.

12.Explanation

Failure in Integration occurs when different components do not work together properly.

Analysis

1. Identify **interface mismatches**.
2. Check for **data inconsistency**.
3. Identify **incompatible components**.
4. Perform **integration testing** and verify interfaces.
5. Ensure all components work together as **one reliable system**.


12. Concept Map
Failure in Integration
Causes
• Interface mismatch
• Data inconsistency
• Incompatible components
How to Fix
• Integration testing
• Verify interfaces
• Resolve compatibility issues
Result
• Detect integration errors
• Ensure smooth working
• Reliable system
Goal
All components work together as one system